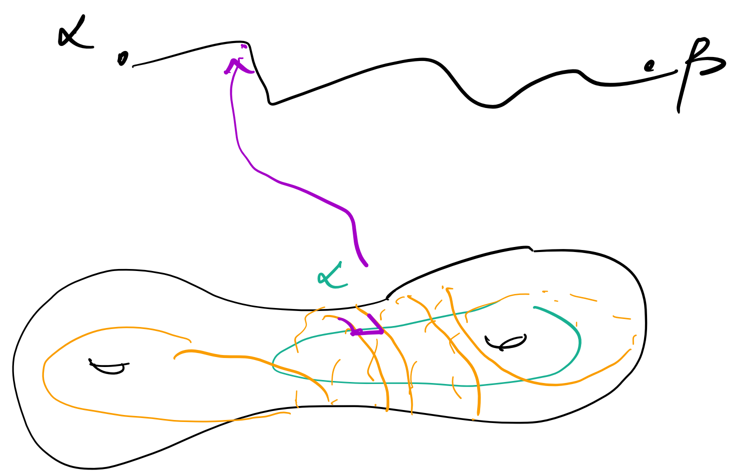
Why is the BGIT true?

what does a geodesic look like in the curve graph?



surgeries: cut and remove parts of β and lay them along α. So just add more pieces of β to α.

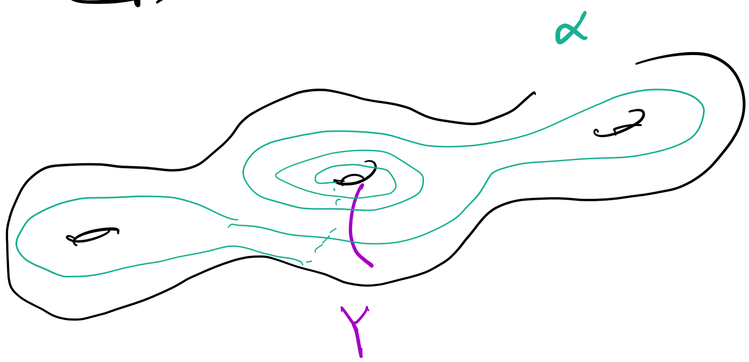
BGIT says: geod $\gamma = (\alpha_1 \cdots \alpha_n)$ in $\mathcal{C}(S)$.

if $i(\alpha_i, \partial Y) > 0 \quad \forall i$ then

$\operatorname{diam} \pi_{\mathcal{C}(Y)}(\gamma)$ unifly bndd.

Conceptually: α_1, α_n have lots of intersection w/ ∂Y, then interchanging pieces of them is likely to preserve high intersection.

e.g.: $d(\partial Y, \alpha) \gg 0$

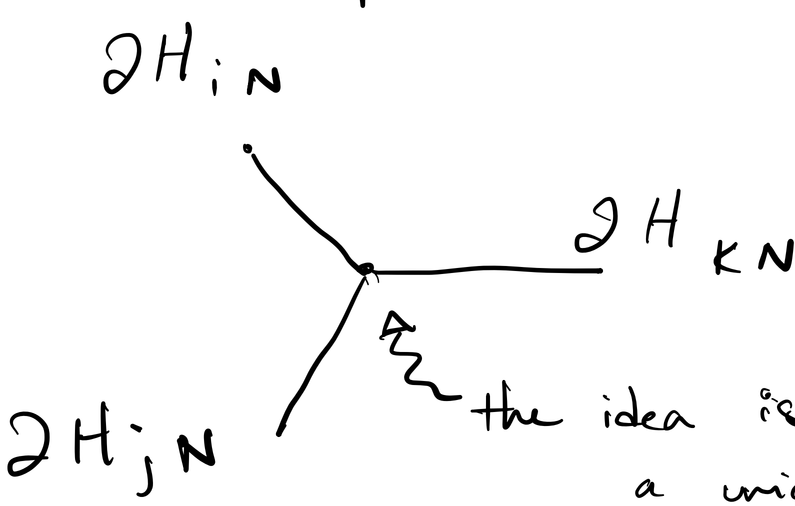


g fully supported on Y (in this case, a Dehn twist)

choose $H_k < \mathrm{Stab}(g^k \alpha)$.

Prop 9.1
for N large, $\{H_{kN}\}$ is D-misaligned.

why? want: tripod in $\mathcal{C}(S)$



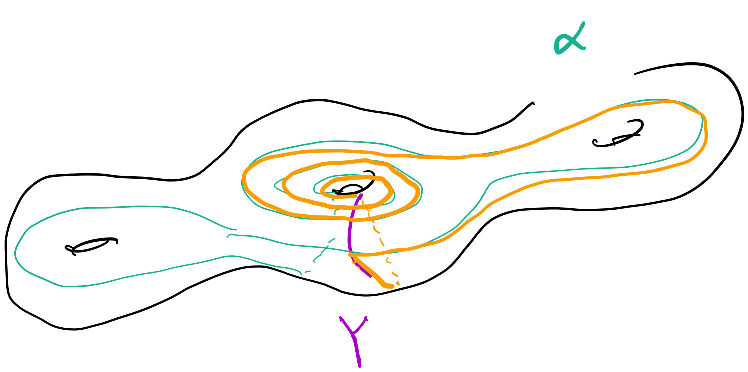
the idea is ∂Y will be a unicorn pt / centroid

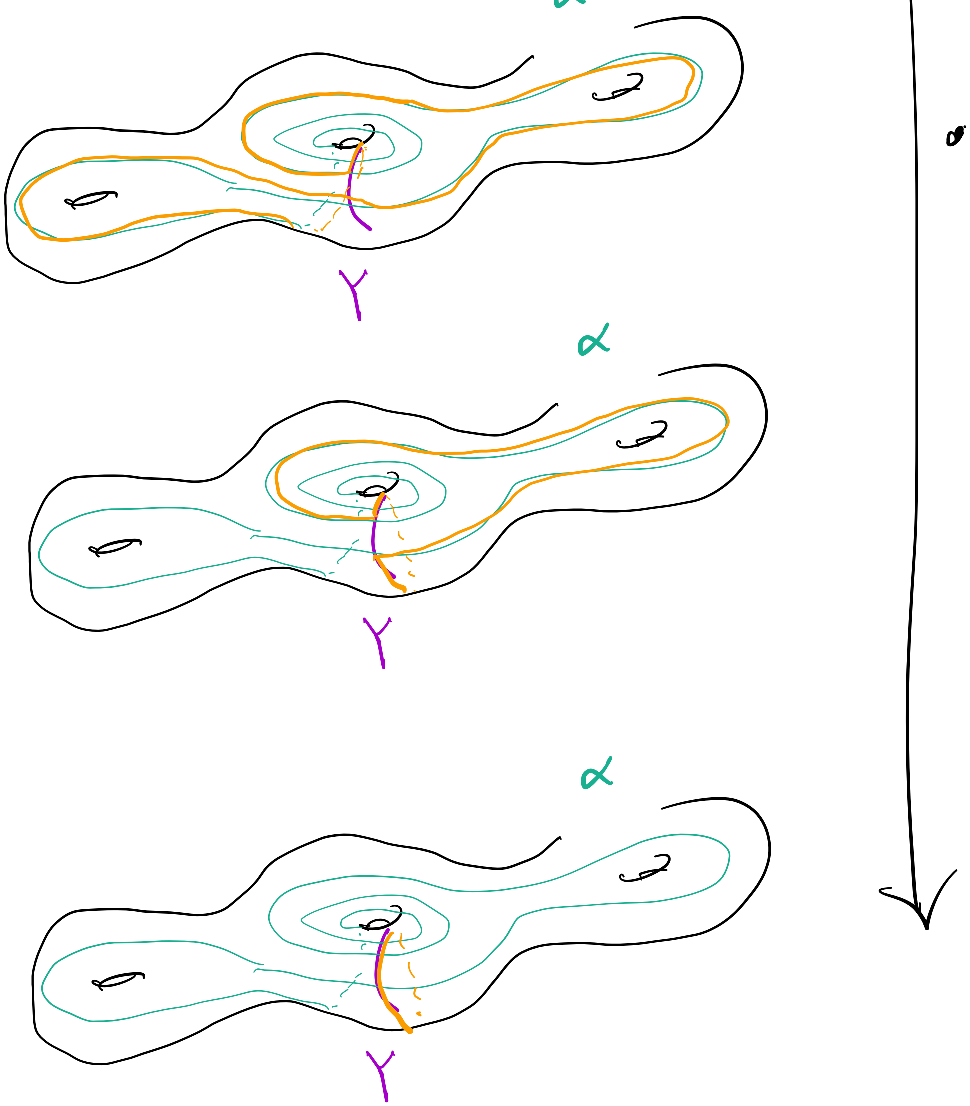
since ∂H_k is a multicurve either containing or disj't from $g^k\alpha$, it suffices to prove the above configuration for the $g^k\alpha$.

b/c g twists along ∂Y, $d(g^k\alpha, \partial Y) \gg 0$

also (for N suffice large).

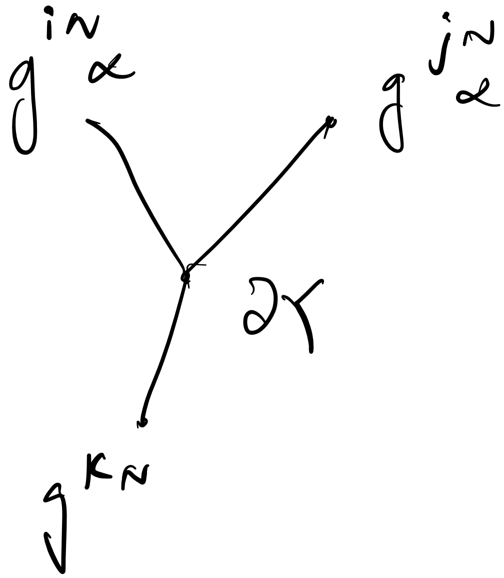
to go from $g^k\alpha$ to $g^{k'}\alpha$ you can undo by a surgery of $g^k\alpha$ along ∂Y until all you see is ∂Y, then rebuilding to get to $g^{k'}\alpha$





- this path is geodesic

- and undoing this path is geodesic
- similar by replacing α w/ $g^k \alpha$
- so $\exists$ geodesic $g^k \alpha$ to $g^{k'} \alpha$ passes near/through $\partial \gamma$.
- same for all of them! Thus,



in $\mathcal{C}(S)$

Thanks to Kasra Rafi for explaining BGIT to me